Project Title: End-to-End Hardware Implementation of a Neural Network

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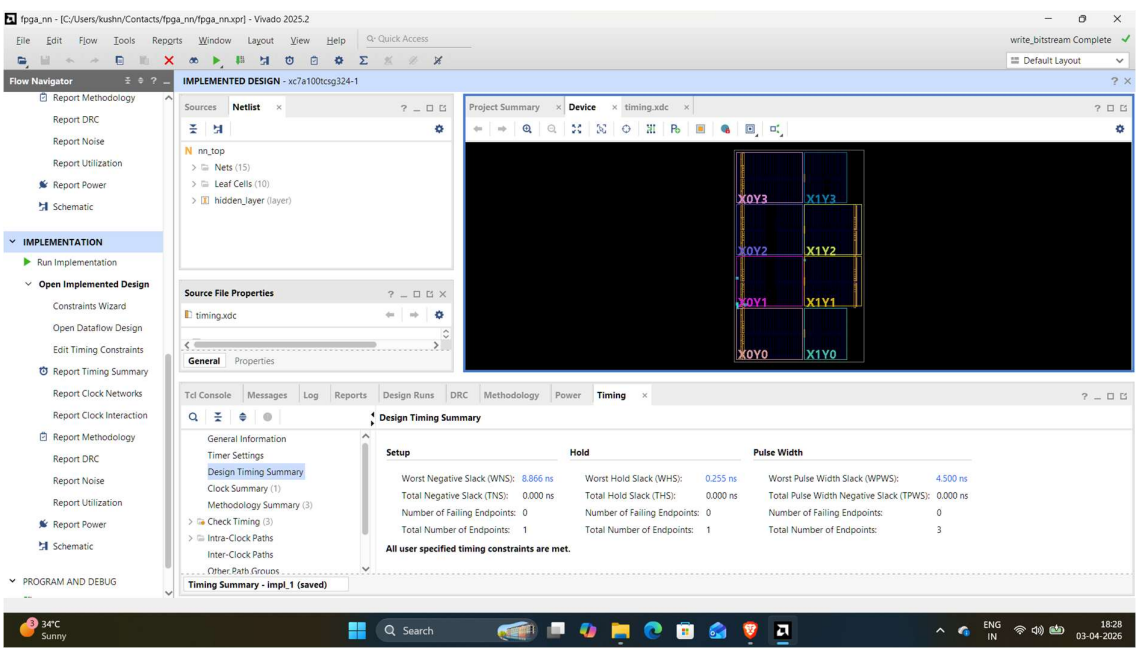
1. Project Overview This project demonstrates the successful training, extraction, and hardware synthesis of an Artificial Neural Network (ANN). A 4-8-3 Multi-Layer Perceptron was trained in Python to classify the Iris dataset. The floating-point weights were converted to Q8 Fixed-Point format (.mem files) and loaded into a custom Verilog architecture.

2. Timing Analysis The design was constrained targeting the xc7a100tcsg324-1 (Nexys A7-100T) FPGA. The timing report confirms the design met all constraints without any violations, ensuring stable clock propagation.

- **Worst Negative Slack (WNS):** 8.866 ns
- **Total Negative Slack (TNS):** 0.000 ns
- **Worst Hold Slack (WHS):** 0.255 ns
- **Total Hold Slack (THS):** 0.000 ns

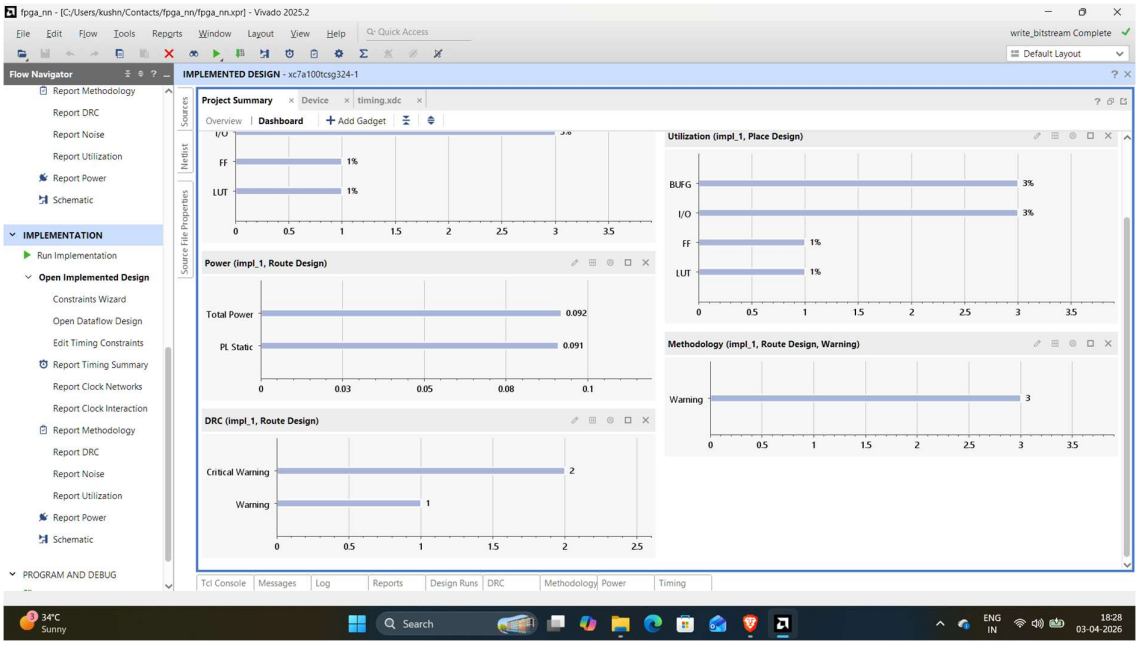


(Screenshot 1: Vivado Design Timing Summary)

3. Resource and Power Utilization The design is highly optimized, utilizing minimal hardware resources on the Artix-7 chip, leaving ample room for future network scaling and parallelization. Total on-chip power consumption is also minimal.

- **LUT Utilization:** 1%
- **Flip-Flop (FF) Utilization:** 1%

- **BUFG:** 3%
- **Total On-Chip Power:** 0.092 W



(Screenshot 2: Vivado Project Summary and Utilization Dashboard)

4. Conclusion The bitstream was successfully generated without critical timing failures. The WNS of 8.866 ns indicates the critical path logic is highly efficient, proving the fixed-point Verilog architecture is robust and ready for physical hardware deployment.